TRANSPORTATION PROBLEM

1.

		DESTINATION			
		CAR	DOLL	BOARDGAME	SUPPLY
	T1	20	20	20	45
SOURCE	T2	25	25	M	60
	T3	12	M	10	40
	DUMMY	0	0	0	10
	DEMAND	50	65	40	

DECISION VARIABLE: X_{ij} $\quad i \in \{1, 2, 3\}$ TEAM

$j \in \{C, D, B\}$ TOYS

= amount of toys assigned to each team

2.

	1	2	3	SUPPLY	u_i
1	9 / 8	10 / 4 −	7 / −2 E +	12	0
2	3 / 2	2 / 5 +	1 / 2 − L	7	−8
3	6 / 4	3 / 0	2 / 11	11	−7
DEMAND	8	9	13		
v_j	9	10	9		

x_{13} ENTERING

x_{23} LEAVING

$Z = 8 \cdot 9 + 10 \cdot 4 + 2 \cdot 5 + 1 \cdot 2 + 2 \cdot 11$
$= 146$

	1	2	3	SUPPLY	u_i
1	9 / 8	10 / 2	7 / 2	12	0
2	3 / 2	2 / 7	1 / 2	7	−8
3	6 / 2	3 / −2	2 / 11	11	−5
DEMAND	8	9	13		
v_j	9	10	7		

$c_{ij} - u_i - v_j$

$2 - 7 = 4$

$3 - 10 + 5 = -2$

$Z = 8 \cdot 9 + 10 \cdot 2 + 7 \cdot 2 + 2 \cdot 7 + 2 \cdot 11$
$= 142$

$x_{11} = 8 \quad x_{12} = 2 \quad x_{13} = 2$

$x_{22} = 7 \quad x_{33} = 11$

the solution is not optimal because there is still a negative coefficient

QDP - DENTAL CLINIC

$$\begin{tikzcd}
0 \arrow[r, bend left, "6"] & 1 \arrow[l, bend left, "3"] \arrow[r, bend left, "6"] & 2 \arrow[l, bend left, "6"] \arrow[r, bend left, "6"] & 3 \arrow[l, bend left, "9"] \arrow[r, bend left, "6"] & 4 \arrow[l, bend left, "9"]
\end{tikzcd}$$

1. IN = OUT

$$\begin{cases} 3\pi_1 = 6\pi_0 \\ 6\pi_2 = 6\pi_1 \\ 9\pi_3 = 6\pi_2 \\ 9\pi_4 = 6\pi_3 \\ \pi_0 + \pi_1 + \pi_2 + \pi_3 + \pi_4 = 1 \end{cases}$$

$$\pi_1 = 2\pi_0$$

$$\pi_2 = \pi_1 = 2\pi_0$$

$$\pi_3 = \frac{6}{9}\pi_2 = \frac{12}{9}\pi_0 = \frac{4}{3}\pi_0$$

$$\pi_4 = \frac{6}{9}\pi_3 = \frac{2}{3}\cdot\frac{4}{3}\pi_0 = \frac{8}{9}\pi_0$$

$$\pi_0 + 2\pi_0 + 2\pi_0 + \frac{4}{3}\pi_0 + \frac{8}{9}\pi_0 = 1$$

$$(9 + 18 + 18 + 12 + 8)\pi_0 = 9$$

$$\pi_0 = \frac{9}{65}$$

$$\pi = \left(\frac{9}{65}, \frac{18}{65}, \frac{18}{65}, \frac{12}{65}, \frac{8}{65}\right)$$

$$\# \text{PATIENTS} = 0\cdot\frac{9}{65} + 1\cdot\frac{18}{65} + 2\cdot\frac{18}{65} + 3\cdot\frac{12}{65} + 4\cdot\frac{8}{65}$$

$$= \frac{(18 + 36 + 36 + 32)}{65} = \frac{122}{65} \sim 1{,}87$$

2. E_x = time to go from state x to state 3

$$\begin{cases} E_0 = \frac{1}{6} + E_1 \\ E_1 = \frac{1}{9} + \frac{3}{9}E_0 + \frac{6}{9}E_2 \\ E_2 = \frac{1}{12} + \frac{6}{12}E_1 + \frac{6}{12}E_3 \\ E_3 = 0 \end{cases}$$ (HOURS)

$$\begin{cases} E_0 = 10 + E_1 \\ E_1 = \frac{20}{3} + \frac{1}{3}E_0 + \frac{2}{3}E_2 \\ E_2 = 5 + \frac{1}{2}E_1 + \frac{1}{2}E_3 \\ E_3 = 0 \end{cases}$$ (MINUTES)

$$E_2 = \frac{1}{12} + \frac{1}{2}E_1$$

$$E_1 = \frac{1}{9} + \frac{1}{3}E_0 + \frac{2}{3}\left(\frac{1}{12} + \frac{1}{2}E_1\right)$$

$$= \frac{1}{9} + \frac{1}{3}E_0 + \frac{1}{18} + \frac{1}{3}E_1$$

$$\frac{2}{3}E_1 = \frac{1}{6} + \frac{1}{3}E_0$$

$$E_1 = \frac{3}{2}\left(\frac{1}{6} + \frac{1}{3}E_0\right)$$

$$= \frac{1}{4} + \frac{1}{2}E_0$$

$$E_0 = \frac{1}{6} + \frac{1}{4} + \frac{1}{2}E_0$$

$$\frac{1}{2}E_0 = \frac{2+3}{12} = \frac{5}{12}$$

$E_0 = \frac{5}{6}$ hours (50 minutes)

SIMPLEX METHOD

$$\max \quad 3000\, x_L + 5000\, x_P$$

$$x_L + x_P \le 12$$

$$7 x_L + \quad \le 70$$

$$3 x_P \le 18$$

$$10 x_L + 20 x_P \le 160$$

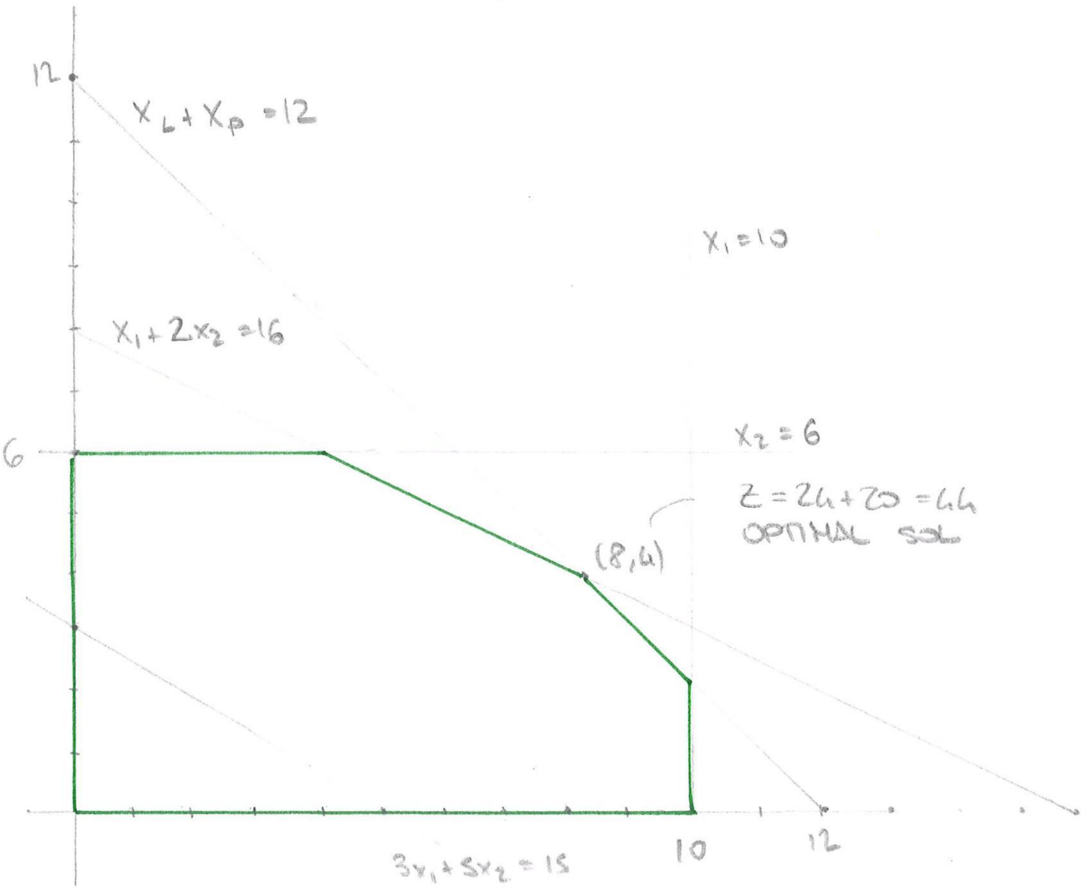
$$\max \quad 3x_1 + 5x_2$$

$$x_1 + x_2 \le 12$$

$$x_1 \le 10$$

$$x_2 \le 6$$

$$x_1 + 2x_2 \le 16$$



$$\max \quad 3000x_1 + 5000x_2$$
$$x_1 + x_2 \le 12$$
$$7x_1 \le 70$$
$$3x_2 \le 18$$
$$10x_1 + 20x_2 \le 160$$

$$\max \quad 3x_1 + 5x_2$$
$$x_1 + x_2 \le 12$$
$$x_1 \le 10$$
$$x_2 \le 6$$
$$x_1 + 2x_2 \le 16$$

$$\max \quad z$$
$$z - 3x_1 - 5x_2 = 0$$
$$x_1 + x_2 + x_3 = 12$$
$$x_1 + x_4 = 10$$
$$x_2 + x_5 = 6$$
$$x_1 + 2x_2 + x_6 = 16$$

ITER	BV	z	x_1	x_2	x_3	x_4	x_5	x_6	RHS	
1	z	1	-3	-5	0	0	0	0	0	
	x_3	0	1	1	1	0	0	0	12	12/1 = 12
	x_4	0	1	0	0	1	0	0	10	—
	x_5	0	0	1	0	0	1	0	6	6/1 = 6
	x_6	0	1	2	0	0	0	1	16	16/2 = 8
2	z	1	-3	0	0	0	5	0	30	
	x_3	0	1	0	1	0	-1	0	6	6/1 = 6
	x_4	0	1	0	0	1	0	0	10	10/1 = 10
	x_2	0	0	1	0	0	1	0	6	—
	x_6	0	1	0	0	0	-2	1	4	4/1 = 4

ITER	BV	Z	x_1	x_2	x_3	x_4	x_5	x_6	RHS	
3	Z	1	0	0	0	0	-1	3	42	
	x_3	0	0	0	1	0	1	-1	2	2/1 = 2
	x_4	0	0	0	0	1	2	-1	6	6/2 = 3
	x_2	0	0	1	0	0	1	0	6	6/1 = 6
	x_1	0	1	0	0	0	-2	1	4	—
4	Z	1	0	0	1	0	0	2	44	
	x_5	0	0	0	0	0	1	-1	2	
	x_4	0	0	0	-2	1	0	1	2	
	x_2	0	0	1	-1	0	0	1	4	
	x_1	0	1	0	2	0	0	-1	8	

OPTIMAL SOL: $x_1 = 8$ $z = 44$

$x_2 = 4$

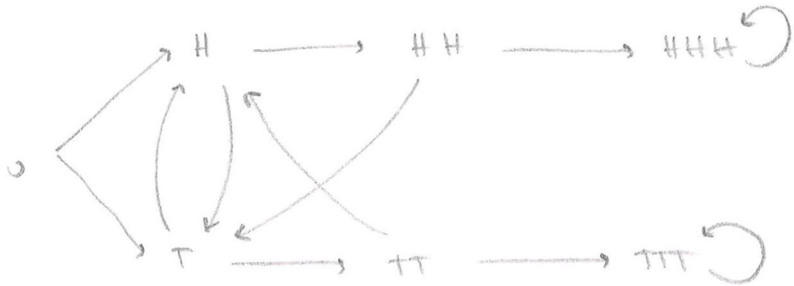
$x_3 = 0$

$x_4 = 2$

$x_5 = 2$

$x_6 = 0$

COIN FLIPPING



(P_{HHH} finit)

P_x of reaching HHH before TTT

$$\begin{cases} P_0 = \frac{1}{2} P_H + \frac{1}{2} P_T \\ P_H = \frac{1}{2} P_T + \frac{1}{2} P_{HH} \\ P_T = \frac{1}{2} P_H + \frac{1}{2} P_{TT} \\ P_{HH} = \frac{1}{2} P_{HHH} + \frac{1}{2} P_T \\ P_{TT} = \frac{1}{2} P_{TTT} + \frac{1}{2} P_H \\ P_{HHH} = 1 \\ P_{TTT} = 0 \end{cases}$$

$$P_{TT} = \frac{1}{2} P_H$$

$$P_{HH} = \frac{1}{2} + \frac{1}{2} P_T$$

$$P_T = \frac{1}{2} P_H + \frac{1}{4} P_H = \frac{3}{4} P_H$$

$$P_H = \frac{3}{8} P_H + \frac{1}{2}\left(\frac{1}{2} + \frac{1}{2}\left(\frac{3}{4} P_H\right)\right)$$

$$= \frac{3}{8} P_H + \frac{1}{4} + \frac{3}{16} P_H$$

$$1 - \frac{3}{8} - \frac{3}{16} = \frac{16 - 6 - 3}{16} = \frac{7}{16}$$

$$\frac{7}{16} P_H = \frac{1}{4} \qquad P_H = \frac{16}{7} \cdot \frac{1}{4} = \frac{4}{7} \qquad P_T = \frac{3}{7}$$

$$P_0 = \frac{2}{7} + \frac{3}{14} = \frac{1}{2}$$

2. $S \xrightarrow[2]{} H \xrightarrow[4]{} HH \xrightarrow[8]{} HHH \qquad 14$

or

ALL TRANSITION PROB are $\frac{1}{2}$

E_x = number of coin flips to go from x to HHH

$$\begin{cases} E_S = 1 + \frac{1}{2}E_H + \frac{1}{2}E_S \\ E_H = 1 + \frac{1}{2}E_{HH} + \frac{1}{2}E_S \\ E_{HH} = 1 + \frac{1}{2}E_{HHH} + \frac{1}{2}E_S \\ E_{HHH} = 0 \end{cases}$$

$$E_{HH} = 1 + \frac{1}{2}E_S$$

$$E_H = 1 + \frac{1}{2}\left(1 + \frac{1}{2}E_S\right) + \frac{1}{2}E_S$$

$$= 1 + \frac{1}{2} + \frac{1}{4}E_S + \frac{1}{2}E_S$$

$$= \frac{3}{2} + \frac{3}{4}E_S$$

$$E_S = 1 + \frac{1}{2}\left(\frac{3}{2} + \frac{3}{4}E_S\right) + \frac{1}{2}E_S$$

$$= 1 + \frac{3}{4} + \frac{3}{8}E_S + \frac{1}{2}E_S$$

$$\left(1 - \frac{3}{8} - \frac{1}{2}\right)E_S = \frac{7}{4}$$

$$\frac{8-3-4}{8}E_S = \frac{7}{4}$$

$$\frac{1}{8}E_S = \frac{7}{4} \qquad E_S = 14$$

GAME THEORY

$$\begin{array}{c|cc|c} & L & R & \\ \hline T & 0,-1 & 2,0 & p \\ B & 1,2 & 0,1 & 1-p \\ \hline & q & 1-q & \end{array}$$

max profit

$R_1(L) = B$

$R_1(R) = T$

$R_2(T) = R$

$R_2(B) = L$

PURE NE : (B,L) with value $(1,2)$

(T,R) $(2,0)$

EXPECTED PAYOFF P1 :

$0q + 2(1-q)$ T

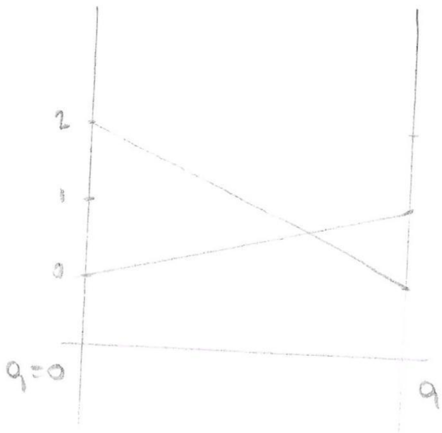
$q + 0(1-q)$ B

$2(1-q) = q$

$2 - 2q = q$

$q = \frac{2}{3}$

$v = \frac{2}{3}$



EXPECTED PAYOFF P2 :

$-1p + 2(1-p)$ L

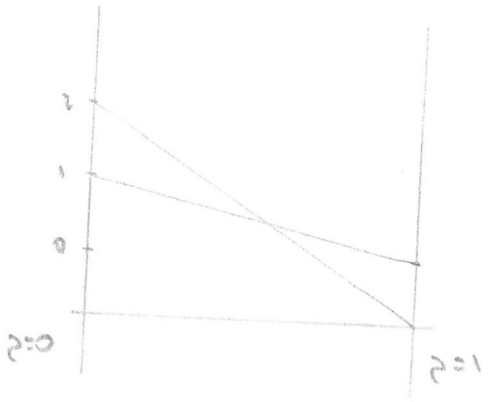
$0p + 1(1-p)$ R

$-p + 2(1-p) = 1-p$

$2 - 2p = 1$

$p = \frac{1}{2}$

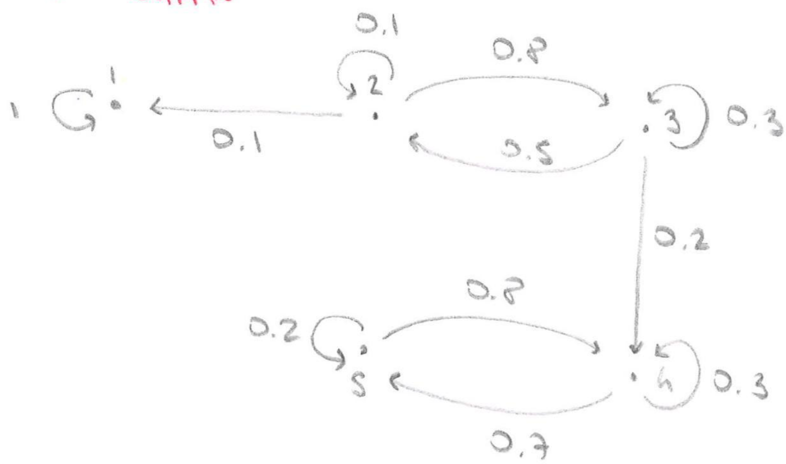
$v = \frac{1}{2}$



MIXED NE : $(\frac{1}{2}, \frac{2}{3})$ with value $(\frac{1}{2}, \frac{2}{3})$

MARKOV CHAIN

(AACHEN)



1. P_x = prob of ever arriving in state 1 starting at x

$$\begin{cases} P_1 = 1 \\ P_2 = 0.1 P_1 + 0.1 P_2 + 0.8 P_3 \\ P_3 = 0.5 P_2 + 0.3 P_3 + 0.2 P_4 \\ P_4 = P_5 = 0 \end{cases}$$

$P_3 = 0.5 P_2 + 0.3 P_3$

$P_3 = \frac{5}{7} P_2$

$P_2 = 0.1 + 0.1 P_2 + \frac{8}{10} \cdot \frac{5}{7} P_2$

$= 0.1 + 0.1 P_2 + \frac{4}{7} P_2$

$\left(1 - \frac{1}{10} - \frac{4}{7}\right) P_2 = \frac{1}{10}$

$\frac{70 - 7 - 40}{70} = \frac{23}{70}$

$P_2 = \frac{70}{23} \cdot \frac{1}{10} = \frac{7}{23}$

$P_3 = \frac{5}{7} \cdot \frac{7}{23} = \frac{5}{23}$ prob of arriving in 1 0.217 $\frac{90}{414}$

$1 - \frac{5}{23} = \frac{18}{23}$ prob of arriving in 4

2. $\pi_1 = \frac{5}{23}$

$\pi_2 = \pi_3 = 0$ transient states

$$P_{45} = \begin{bmatrix} 0.3 & 0.7 \\ 0.8 & 0.2 \end{bmatrix}$$

$$\hat{\pi}_{45} = \hat{\pi}_{45} P_{45} \quad \Rightarrow \quad \begin{aligned} \hat{\pi}_4 &= 0.3\hat{\pi}_4 + 0.8\hat{\pi}_5 \\ \hat{\pi}_5 &= 0.7\hat{\pi}_4 + 0.2\hat{\pi}_5 \end{aligned}$$

$$\hat{\pi}_4 = \frac{8}{7}\hat{\pi}_5$$

$$\hat{\pi}_5 + \frac{8}{7}\hat{\pi}_5 = 1$$

$$\hat{\pi}_5 = \frac{7}{15} \qquad \hat{\pi}_4 = \frac{8}{15}$$

$$\pi = \left(\frac{5}{23}, 0, 0, \frac{18}{23}\cdot\frac{8}{15}, \frac{18}{23}\cdot\frac{7}{15}\right)$$
$$= \left(\frac{5}{23}, 0, 0, \frac{48}{115}, \frac{42}{115}\right)$$

3. $$P_T = \begin{bmatrix} 0.1 & 0.8 \\ 0.5 & 0.3 \end{bmatrix}$$

$$S = (I - P_T)^{-1} = \begin{bmatrix} 0.9 & -0.8 \\ -0.5 & 0.7 \end{bmatrix}^{-1}$$

$$= \frac{1}{\frac{9}{10}\cdot\frac{7}{10} - \frac{8}{10}\cdot\frac{5}{10}} \begin{bmatrix} 0.7 & 0.8 \\ 0.5 & 0.9 \end{bmatrix}$$

$$= \frac{100}{23}\begin{bmatrix} \frac{7}{10} & \frac{8}{10} \\ \frac{5}{10} & \frac{9}{10} \end{bmatrix} = \begin{bmatrix} \frac{70}{23} & \frac{80}{23} \\ \frac{50}{23} & \frac{90}{23} \end{bmatrix}$$

starting in 2 we will stay $\frac{70}{23}$ steps in state 2

$\frac{80}{23}$ 3